

Troubleshooting

Real errors people hit with this toolkit, what causes them, and how to fix each one.

Earth Engine

ValueError: Set EE_PROJECT to your own Google Cloud project...

Working as intended. The toolkit ships with no project ID, because Earth Engine projects are tied to one Google account and no shared default could work for everyone. Put your own Project ID in the quotes (notebooks) or the sidebar field (app). See the Getting Started guide.

EEException: Cannot authenticate: Invalid request.

Google retired the "out-of-band" OAuth flow, where you copied an authorization code from a browser and pasted it back. Any code still calling a bare `ee.Authenticate()` that falls back to that flow now fails this way. This toolkit calls `ee.Authenticate(auth_mode="localhost")`, which opens your browser and completes automatically. If you still see this, you're running an older copy — update it.

EEException: Please authorize access to your Earth Engine account...

You have no cached credentials yet. Run the setup cell (or click Connect in the app) and complete the browser sign-in. If it persists, delete the cached credentials file and retry — on Windows it lives under `%USERPROFILE%\config\earthengine\`.

Permission denied / project not registered

The Project ID is valid but that project isn't registered for Earth Engine, or you signed in with a different Google account than the one you registered. Re-register at `code.earthengine.google.com/register` and make sure the account matches. Also check you used the Project **ID**, not the display name.

Downloads (NASA Earthdata / ASF)

ASFAuthenticationError: Failed to log in with provided credentials (401)

Almost always **not** a wrong password. ASF requires you to authorize their application against your Earthdata account before their API will accept you. Go to `urs.earthdata.nasa.gov/profile/edit_applications`, find **"Alaska Satellite Facility Data Access"** under Authorized Apps, approve it, and retry. If it still fails, confirm you're using your Earthdata **username**, not the email address you registered with — they are often different.

A download stops partway and won't resume

Safe to just re-run. Downloads write to a temporary `.part` file and are only renamed once complete, so an interrupted transfer is never mistaken for a finished one. Completed files are skipped.

Search results

A sensor returns 0 scenes

Usually correct, not a bug. The three common reasons:

Sensor	Why it can legitimately return nothing
Sentinel-1	Acquisition is tasked by ESA, not automatic. The satellite flies over on schedule, but whether it collects your AOI is a planning decision that changes over time.
Sentinel-6	Ocean-only altimetry. A land AOI will always return zero.
NISAR	Early-mission/commissioning phase, so coverage is still sparse and provisional.

■ **Tip** — To confirm independently, search the same AOI and dates in the mission's own web tool — ASF Vertex (search.asf.alaska.edu) for Sentinel-1/NISAR, or NASA Earthdata Search (search.earthdata.nasa.gov) for anything CMR-hosted. If those show nothing either, the data genuinely isn't there.

MODIS footprints cover the whole world on the map

Expected. MODIS products here are global daily/composite mosaics, not swath-limited scenes, so their footprint is the globe and `aoi_cov_pct` is always ~100%. The coverage filter isn't meaningful for MODIS — filter by date and product instead.

Notebooks and the app

The Jupyter kernel crashes in the search or forecast notebook

Usually memory pressure from large embedded map widgets accumulating in saved output. Restart the kernel and clear outputs (`jupyter nbconvert --clear-output --inplace *.ipynb`), then re-run from the top.

Map tiles show "API KEY REQUIRED" watermarks

CartoDB's basemap tiles now require an API key. The toolkit uses OpenStreetMap, which doesn't. If you see this, you're on an older copy — update it.

"Connect to Earth Engine" in the app does nothing useful

Enter your Project ID in the sidebar field first; the button refuses to run without one.

Installation (Windows)

venv creation fails: ensurepip ... returned non-zero exit status 1

Windows' 260-character path limit. Creating a virtual environment inside a deeply nested folder — especially a OneDrive-synced Desktop path — silently breaks pip bootstrapping. Create the

environment somewhere short instead:

```
python -m venv C:\venvs\sds
C:\venvs\sds\Scripts\activate
pip install -r requirements.txt
```

■ **Tip** — Either enable long-path support in Windows, or keep virtual environments out of OneDrive-synced folders entirely — sync can also lock files mid-install.

GitHub sync

401 Client Error: Unauthorized (api.github.com)

GitHub rejected the token itself — this is not a scope or permission problem (those return 403). Check that the token hasn't expired or been revoked, that no stray whitespace was pasted with it, and that you're using a Personal Access Token rather than your account password, which GitHub stopped accepting in 2021.

Push rejected as non-fast-forward

The repository has commits you don't have locally — often from editing a file directly on github.com. Run `git pull --rebase origin main`, resolve anything that conflicts, then push again. Avoid force-pushing unless you're certain you want to discard what's on the remote.